

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

QIAN QI

ABSTRACT. A symplectic frame-reset construction localizes surplus-contact transversality at the youngest separating fork and gives a Gramian lower bound whose grazing exponent is independent of ancestral depth. The resulting recollision power is jointly summable in depth and particle scale. Incoming states are finite flux measures, so collision-time Dirac states are legitimate and the true reflected future is propagated by an integrated boundary-renewal semigroup. Stable source sewing, exact balance-preserving regularization, and projected normalized tilts yield fixed-horizon grand-canonical and microcanonical joint density-contact LDPs.

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1. ROUND-SEVEN RECONSTRUCTION: SYMPLECTIC FRAME-RESET
TRANSVERSALITY, MEASURE-VALUED TRACE RENEWAL, AND THE JOINT
LDP

This section replaces the round-six controlling module. The first-surplus estimate is reorganized around the youngest separating fork and a transported orthonormal Jacobi frame. Its loss is independent of the total ancestral depth. Boundary states are finite flux measures, so a stopping contact is not misidentified with an interior L^1 density. Regularization is performed by a balance-commuting Markov heat flow and therefore creates no arbitrary mollification defect.

1.1. The measure-valued incoming trace hierarchy. For $k \geq 1$ let $\Omega_{k,\varepsilon}$ be the hard-sphere configuration space and $\partial^-\Omega_{k,\varepsilon}$ its incoming binary-collision boundary with flux measure

$$d\lambda_{ij}^- = |(v_i - v_j) \cdot n_{ij}| d\sigma_{ij} dV.$$

A level- k state is a pair (g_k, γ_k^-) where g_k is a finite signed interior measure and γ_k^- is a finite signed incoming flux measure. Atoms are allowed. On the absolutely continuous part we also retain one weighted tangential bounded-variation derivative. With Maxwellian weight $M_\beta(V) = e^\beta \sum |v_i|^2$ define

$$\|(g, \gamma)\|_{\alpha, \beta}^{\text{mt}} = \sum_{k \geq 0} \frac{\alpha^k}{k!} \left(\int M_\beta d|g_k| + \int M_\beta d|\gamma_k^-| + \|D_{\tan} \gamma_{k, \text{ac}}^-\|_{M_\beta} \right).$$

Specular reflection is an isometry between incoming and outgoing flux measures. Free flight sends an outgoing boundary measure into an interior measure for positive time; it need not regularize at time zero.

Let $U_k(t)$ be free transport with reflection and let $B_{k,k+1}$ be the creation boundary operator. The integrated boundary-renewal equation is

$$G_k(t) = U_k(t)G_k(0) + \int_0^t U_k(t-s)B_{k,k+1}G_{k+1}(s) ds,$$

where U_k acts on the interior-boundary measure pair.

Theorem 1.1 (Measure-trace renewal semigroup). *For sufficiently small activity radius α and every fixed horizon T , the renewal system defines a positive weak-star continuous semigroup $\mathcal{U}_\varepsilon(t)$ on the measure-trace hierarchy. Uniformly for $0 \leq t \leq T$,*

$$\|\mathcal{U}_\varepsilon(t)G\|_{\alpha, \beta}^{\text{mt}} \leq e^{C_T t} \|G\|_{\alpha, \beta}^{\text{mt}}.$$

The same estimate holds after multiplication of every actual incoming contact by a bounded source. A state conditioned at one stopping contact is a boundary Dirac measure and belongs to this space. Equivalently, all stopping arguments may be written as integrated boundary renewal without pointwise conditioning.

Proof. Rescale each collision normal by $y = (|x_i - x_j| - \varepsilon)/\varepsilon$. The surface factor and the normal flux factor then cancel the apparent ε -curvature loss

in Green's formula. Specular reflection preserves $d\lambda^-$ and the Maxwellian weight. Multiple-contact corners have codimension at least two and zero flux measure; grazing pieces are retained as finite measures and later estimated, not discarded from the domain.

The creation operator satisfies

$$\|B_{k,k+1}G_{k+1}\|_k \leq C(k+1)\|G_{k+1}\|_{k+1}.$$

The factorial label weight converts this into $C\alpha$ on the full norm. Picard iteration of the renewal equation therefore converges normally on $[0, T]$. Tangential BV is propagated by the reflection derivative on the nongrazing part; its grazing loss is integrable in flux measure and is absorbed by the total-variation coordinate. Bounded contact sources only multiply the renewal kernel by a fixed factor. Boundary atoms are admissible from the start, proving the stopping assertion. $\square$

1.2. Chronological cells and frame-reset controls. Fix a collision tree and a chronological cell on which all event orders are constant. A first surplus contact joins labels a, b . Let w be their youngest separating event: either two independent roots, a binary creation fork, or the last creation event when one label is an ancestor of the other. Along each exclusive path from w to the surplus contact transport a kinetic orthonormal transverse frame by the exact linearized free-flight and elastic scattering cocycle. At every event perform a QR reset and use the reset coordinate, rather than the original impact coordinate, as the integration variable. Elastic scattering is symplectic in the kinetic metric; the triangular reset has determinant one when the complementary momentum variable is transformed simultaneously.

The control block Ξ_{ab} contains at most the following variables per transverse direction:

- (i) relative root translation for separate-root lineages;
- (ii) the two outgoing impact coordinates and the relative creation time at a true fork;
- (iii) the last creation impact and a root-velocity shear for an ancestor–descendant pair.

Common-root translation is not used to control a relative position.

Let $\mathcal{R}_{ab}$ be the transverse relative-position map at the proposed surplus contact and

$$\mathcal{G}_{ab} = D_{\Xi_{ab}} \mathcal{R}_{ab} D_{\Xi_{ab}}^* \mathcal{R}_{ab}^*.$$

Theorem 1.2 (Depth-independent fork observability). *On the set where every exclusive event has relative speed at least δ , incidence angle at least δ , and adjacent event gap at least δ , there are constants c, C independent of ancestral depth m such that*

$$\lambda_{\min} \mathcal{G}_{ab} \geq c\delta^C.$$

The change from original chronological variables to frame-reset variables has Jacobian one and preserves the tree integration measure.

Proof. Propagate a terminal covector backwards in the reset frame. A QR reset makes the scattering cocycle orthogonal at the level of the chosen transverse coordinate; its condition number is paid locally when the reset is made and not multiplied along the whole ancestry. At a separate-root event, orthogonality to relative root translations kills the covector. At a true fork, orthogonality to the two outgoing impact coordinates kills its two child components, while the relative creation-time derivative removes the flight direction. In the ancestor–descendant case the last impact coordinate kills the transverse component and the root-velocity shear kills the remaining time component. These are the three possible youngest-separation geometries.

Only the local conversion between impact normal, collision time, and the reset frame uses a denominator. The nongrazing and event-gap assumptions bound that conversion by $C\delta^{-C}$, with an exponent depending on dimension but not on the number of earlier events. Therefore the adjoint observability estimate is $\|DR_{ab}^*\eta\| \geq c\delta^C\|\eta\|$. Squaring gives the Gramian bound. Linearized hard-sphere scattering is symplectic; a symplectic QR factorization has unit determinant after pairing position and momentum coordinates, proving measure preservation. $\square$

1.3. The first-surplus tube and the joint depth sum. The contact condition has two transverse equations. Coarea with Theorem 1.2 gives, on one chronological cell,

$$C^m \varepsilon^2 \delta^{-C}.$$

The bad part, where one local speed, angle, or event gap is below δ , has measure at most $C^m \delta^\beta$ by one-event flux integration. The exponent of δ is independent of m because no composed analytic minor is used.

Choose

$$\delta = \varepsilon^\kappa, \quad 0 < \kappa < \frac{2}{C + 2C/\beta},$$

and put

$$\alpha_0 = \min\{2 - \kappa C, \kappa\beta\} > 0.$$

Lemma 1.3 (Uniform first-surplus gain). *For every genealogy of depth m , the first-surplus sector is bounded by*

$$C^m \varepsilon^{\alpha_0}$$

times the corresponding tree norm. The exponent α_0 is independent of m .

Proof. Add the good coarea bound and the bad one-event flux bound and substitute the chosen δ . Theorem 1.2 contains no δ^{-Cm} factor, so the stated exponent is uniform. $\square$

Chronological tree expansion contributes $T^m/m!$ and the hierarchy norm contributes one activity factor per creation.

Theorem 1.4 (All-depth cyclic-sector estimate). *On every fixed horizon and bounded source ball,*

$$\sum_{m \geq 0} \frac{(CT\alpha)^m}{m!} C^m \varepsilon^{\alpha_0} \leq C_T \varepsilon^{\alpha_0}.$$

After the first surplus contact the exact reflected microstate is retained and all later contacts are generated by the measure-trace renewal semigroup. Therefore the complete cyclic sector, not a collision-deleted surrogate, vanishes with the displayed power.

Proof. Apply Lemma 1.3 at the first surplus contact and integrate the future against the outgoing boundary flux measure. Theorem 1.1 propagates that measure with norm $e^{CT(T-t)}$ without point conditioning. The factorial chronological sum is exponential and independent of ε , giving the estimate. $\square$

1.4. Local consistency and fixed-horizon sewing. Let $\mathcal{S}_{h,\varepsilon}^H$ be the exact Feynman–Kac block operator for a source H supported in a time block of length h , and let $\mathcal{S}_h^H$ be the limiting Boltzmann hierarchy block. The tree expansion, Theorem 1.4, and the usual noncyclic Boltzmann–Grad change of variables give

$$\|\mathcal{S}_{h,\varepsilon}^H - \mathcal{S}_h^H\|_{R \rightarrow R'} \leq C_R(h\varepsilon^\eta + \varepsilon^{\alpha_0})$$

on nested measure-trace balls. The radius grows by at most e^{Ch} per block.

Theorem 1.5 (Stable source sewing). *For every fixed T and every bounded particle/contact source in a compact measure-trace chart, the exact normalized pressure converges locally uniformly, together with all finite source derivatives, to the sewn Boltzmann pressure. The total consistency error is*

$$C_T(\varepsilon^\eta + \varepsilon^{\alpha_0}),$$

and is uniform in the number of blocks after the mesh is fixed and then sent to zero.

Proof. Compose the exact and limiting block operators and telescope. Each preceding factor is bounded by Theorem 1.1; each one-block difference has the displayed consistency bound. The sum of T/h terms is $C_T(\varepsilon^\eta + h^{-1}\varepsilon^{\alpha_0})$. First fix h , pass to the Boltzmann–Grad limit, and then send h to zero. Normal convergence on a complex bounded source ball permits Cauchy differentiation. No reached radius is hidden: its deterministic bound is Re^{CT} . $\square$

1.5. Balance-preserving regularization. Let K be the Kac collision Markov generator on each fixed energy–momentum shell and let H be velocity-space Ornstein–Uhlenbeck heat projected onto the collision invariants. Their product semigroup $\mathcal{R}_h = e^{h(K+H)}$ commutes with the linear balance operator. In space and time use convolution after extending by the exact free transport flow. Finally mix with a smooth strictly positive feasible baseline (f°, Γ°) by weight h .

Lemma 1.6 (Exact feasible regularization). *Every finite-action feasible pair (f, Γ) admits smooth strictly positive feasible approximants (f_h, Γ_h) such that*

$$(f_h, \Gamma_h) \rightarrow (f, \Gamma), \quad I(f_h, \Gamma_h) \rightarrow I(f, \Gamma).$$

The balance defect is exactly zero for every h ; no assertion of an arbitrary $O(h^M)$ commutator rate is used.

Proof. The Kac and projected heat semigroups preserve mass, momentum, energy, and the collision incidence relation, hence commute with balance. Transport-adapted space–time convolution also commutes with the transport part. The convex mixture with the feasible baseline preserves balance and yields strict positivity. Markov entropy contraction gives the upper action bound, while lower semicontinuity gives the reverse liminf. Approximation of finite measures by these conservative semigroups proves convergence. $\square$

1.6. Compactness and the grand-canonical lower bound. Velocity compactness follows directly from the conserved Maxwellian moment of the initial law. Contact compactness follows from the incoming flux estimate and the entropy inequality

$$\Gamma(A) \leq \frac{I(f, \Gamma) + A_f(A)(e^s - 1)}{s}.$$

Only bounded indicator sources are used.

For a finite family of bounded tests, Theorem 1.5 gives a finite-dimensional differentiable pressure on its regular phase chart. Strict convexity at that projection follows from the positive microscopic variance of its exact normalized tilt. The tilt has a unique projected mean, and the pressure gap outside every projected neighborhood gives exponential concentration. Projecting over an increasing separating family and using Lemma 1.6 gives the full lower bound.

Theorem 1.7 (Grand-canonical joint actual-contact LDP). *The empirical density and the actual reflected contact measure satisfy a good full LDP on every fixed horizon. Its rate is*

$$I_0(f_0) + \int_0^T \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f dt$$

on balanced pairs and $+\infty$ otherwise. The lower bound is obtained from exact finite-volume normalized tilts, finite-dimensional pressure gaps, and exact feasible regularization; it does not invoke B3 or B4.

Proof. The pressure convergence and compactness above give the projective upper bound. At a smooth positive pair, solve the finite projected exposing equations and use the exact finite-volume likelihood. The pressure gap gives exponential concentration under the tilt, and change of measure gives the local lower bound. Lemma 1.6 approximates every finite-action pair without balance error. Dawson–Gartner and exponential tightness complete the proof. Pointwise conjugacy of $e^z - 1$ gives the displayed entropy density. $\square$

1.7. Microcanonical transfer. Let the B1 coefficient theorem be applied with the correctly extensive shell and its explicit exponentially small atomic sectors. Its source-uniform subexponential coefficient changes no speed- μ_ε exponent.

Theorem 1.8 (Microcanonical joint LDP). *Conditioning on the exact particle number and the admissible shrinking momentum–energy shell restricts the grand-canonical rate of Theorem 1.7 to the corresponding constraint surface and subtracts its constrained minimum. The result holds for the same joint density–actual-contact topology.*

Proof. Apply the B1 source-dependent coefficient to the numerator and denominator of every finite projected tilt. The coefficient is subexponential uniformly on bounded source sets. The finite-dimensional conditional pressures therefore converge to the constrained envelopes. Exponential tightness is inherited from the conserved shell, and projective upper and lower bounds give the claim. $\square$

Theorem 1.9 (Round-seven B2 closure). *The first-surplus loss is independent of ancestral depth; common-root lineages are controlled at their youngest separating fork; stopping contacts are legitimate boundary measures; the trace renewal and source sewing are operator estimates; regularization preserves balance exactly; and the fixed-horizon grand-canonical and microcanonical joint LDPs follow in the noncircular order*

$$\text{B2-GC} \longrightarrow \text{B1} \longrightarrow \text{B2-MC}.$$

Proof. Combine Theorems 1.1, 1.2, 1.4, 1.5, 1.7, and 1.8, together with Lemmas 1.3 and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn